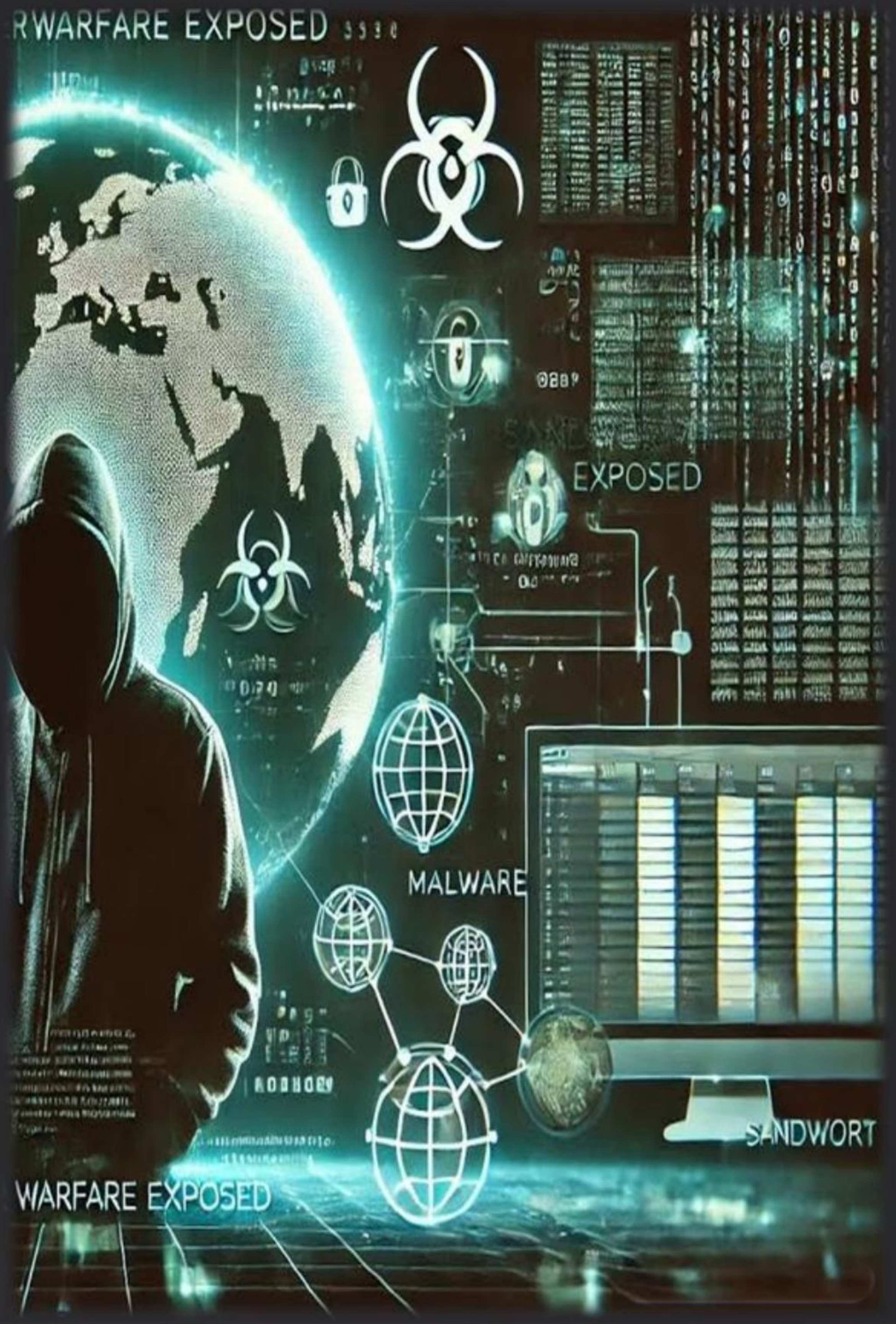
Final project motivation video – predicting cyber attacks

This presentation systematically describes the main components and processes of our cybersecurity threat prediction project, ensuring clarity and a logical flow of information for stakeholders.

Technical interest and product teams.



Why is it important to protect our information from cyber attac

- Protecting privacy and securing personal and business information.
- Preventing loss of funds and digital assets.
- Maintaining the organization's reputation and customer trust.
- Preventing service outages or damage to critical systems.

Data storage

Our project relied on a carefully selected suite of data storage, management, and analytics platforms. This infrastructure was designed for flexibility, scalability, and collaborative efficiency, essential for handling diverse cybersecurity threat data.

1

MongoDB

MongoDB is known for its NoSQL capabilities.
Its, provides flexibility for storage.

2

MySQL

MySQL is used for structured data, managing large datasets, and enabling complex queries and relationship management for analytical consistency.

3

Google Sheets

Google Sheets is used for initial data capture and real-time collaboration.

4

OneDrive

Functioned as our primary cloud storage, providing secure data backup, granular version control, and seamless file synchronization between the team, critical for project continuity and data integrity.

5

Google Collaborate

Our cloud-based development environment Colab enables Python code execution Efficient, extensive data analysis and scalable training of machine learning models with GPU acceleration.

Dashboards

Visualizing cybersecurity threats and insights was critical for effective communication. We integrated different platforms to create dynamic, interactive dashboards and reports that turned complex data into actionable intelligence.

Tableau Public

Used to create dashboards

Interactive, for dynamic visualization of cyber threats.

Looker Studio

Used to generate reports and dashboards

Advanced indicators with direct integration to our data sources, providing deep analytical capabilities and customizable data exploration

personally.

Funnel.io

Enabled visual display of data

Marketing and cybersecurity go hand in hand from multiple different sources, concentrating insights into a holistic view of

Campaign performance and threat landscape.

Power BI

Used to create dashboards

Highly dynamic, allowing measurement Comprehensive performance, strategic decision-making and seamless organizational integration

Threat intelligence.

- Dashboard and indicator table made in Excel

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Data analysis

To deploy and demonstrate our predictive models, we used robust development and interactive application platforms. These tools ensured that our solutions were not only functional but also user-friendly and easily integrated into existing workflows.

Hugging Face (Gradio)

Essential in building intuitive web-based interfaces for our models, allowing users to effortlessly input data and receive real-time cybersecurity threat predictions.

Flask Framework

A lightweight Python web framework used to embed our machine learning models directly into enterprise workflows, making predictive capabilities accessible and practical for end users via API simple.

Matplotlib, Seaborn, Yellowbrick

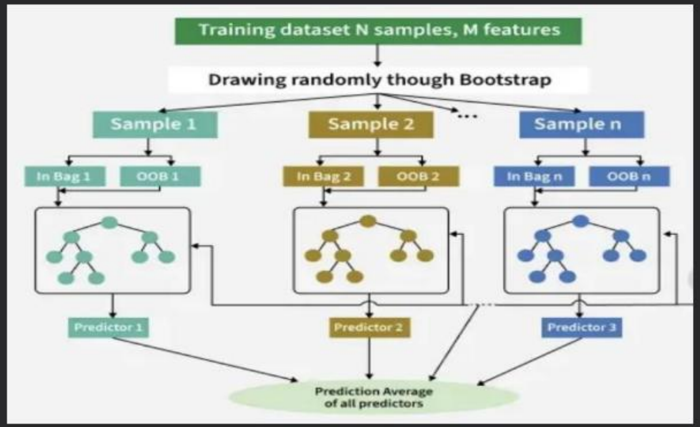
Used to create detailed graphs, charts, and visual representations of data, improving exploratory analysis and model interpretation.

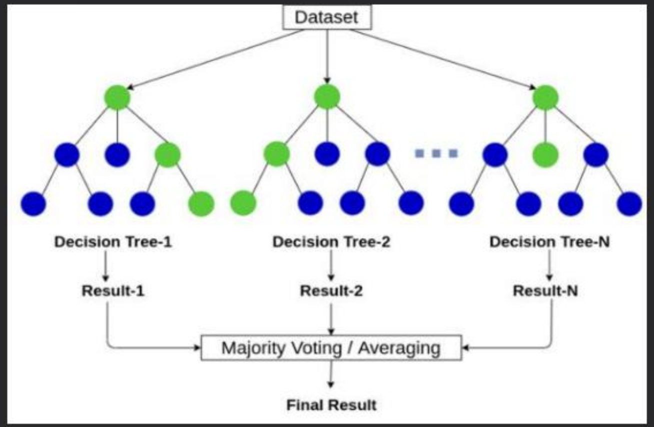
ydata-profiling

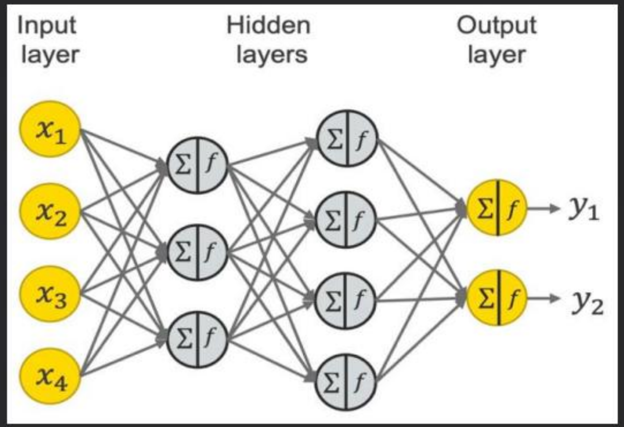
Critical to generating automated and in-depth reports that included statistics, missing value analysis, and insights into our datasets, Accelerate data understanding.

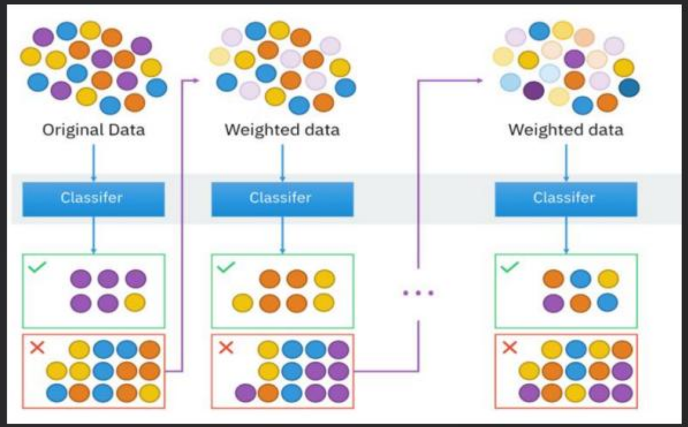
Models

We evaluated several advanced machine learning models to identify the best fit for our prediction task. The basic operation of each model was examined and their performance was accurately evaluated and tested. The dataset was divided into training, validation, and testing sets to ensure robust evaluation and avoid overfitting









CatBoost

Improving the decision tree based on previous trees and error reduction.

Automatic handling of missing values

Random Forest

Combining multiple decision trees for accuracy improved.

Neural Networks

Deep learning model for pattern recognition Complex in data

AdaBoost

Using weak models to create a model

Strengthening and improving the model

Parameter tuning, conducted using cutting-edge tools such as Optuna (automatic tuning of model parameters), was essential to optimizing the performance of each model, ensuring that they would realize the model's potential in predicting cyber events.

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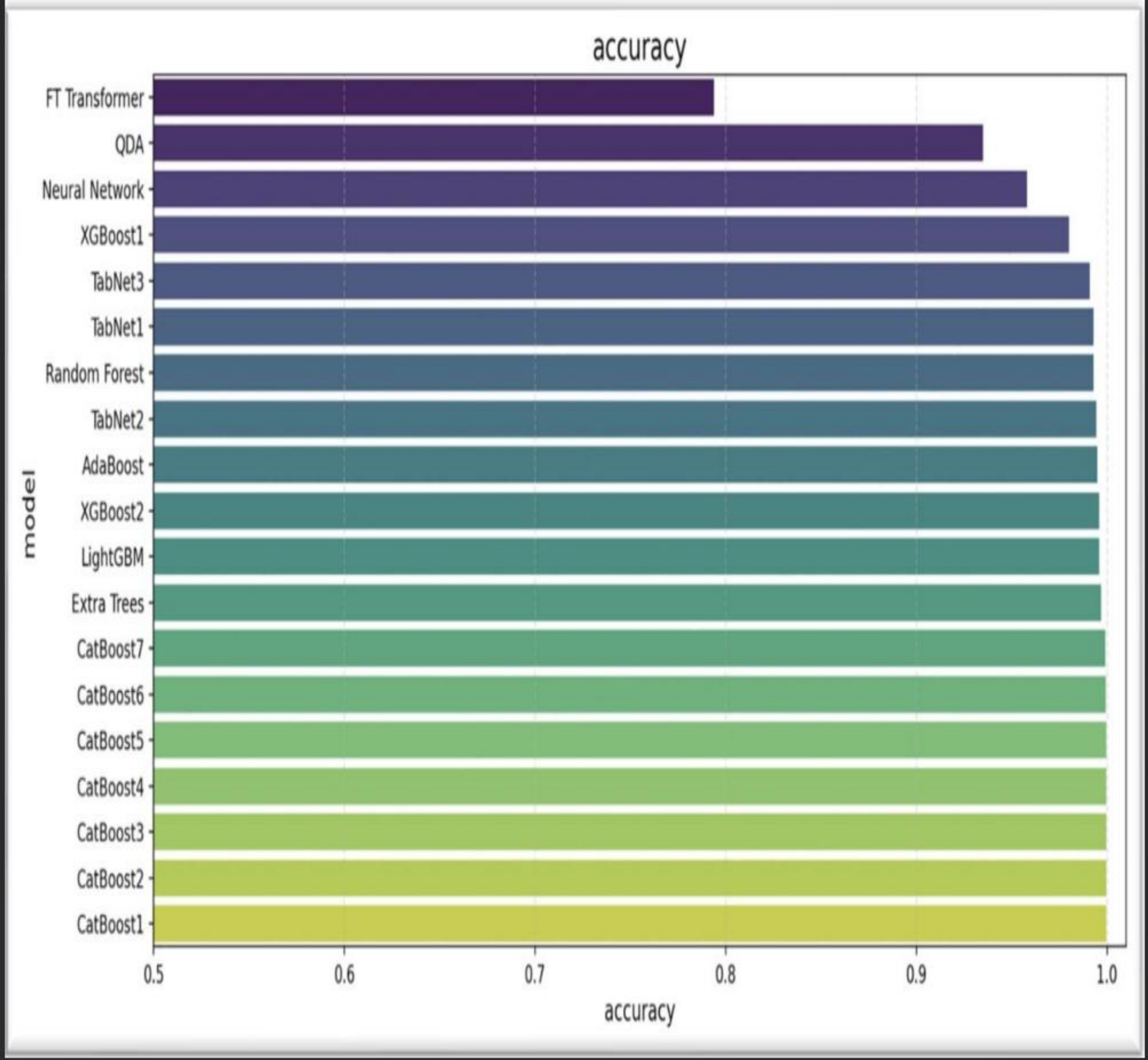
Modeling

Our cybersecurity threat prediction leveraged a diverse portfolio of advanced machine learning models, each selected for its unique strengths in handling complex data and ensuring high prediction accuracy. Parameter optimization was essential to optimize performance.

CatBoost

An ensemble learning method, specifically chosen for its original treatment of category features and resistance to overfitting, which provides high accuracy in predicting threat categories.



XGBoost & LightGBM

Gradient boosting frameworks known for their speed and performance with large datasets, providing highly accurate and efficient threat prediction capabilities.

Extra Trees

An ensemble technique similar to random forest, offering improved generalization capabilities and reduced variance, contributing to more stable and reliable predictions.

TabNet

A modern neural network architecture adapted to tabular data excels in interpretation by identifying and prioritizing key features that influence threat prediction.

Random Forest

A versatile ensemble model, efficient in handling multidimensional data and very powerful for classification and regression tasks, used for basic prediction and feature importance analysis.

Optuna

An automated parameter optimization framework that significantly improved model performance by efficiently searching for optimal configurations across different algorithms.

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Links


Microsoft Type Attack
Microsoft Cyber Attack
Microsoft Attack
Cyber Attack Prediction
tableau
tableau
Looker Studio
OneDrive
mongoDB
Power BI
X
Google Sheets
GitHub
CO


CB
XGB
LGBM
CatBoost
XGBoost
LightGBM
ET
Extra Trees
TabNet
RF
TabNet
RF
Random Forest
OPTUNA
Optuna
ai